AN

MINI PROJECT REPORT

ON

"Titanic Survival Prediction Using Machine Learning"

Submitted in partial fulfillment of

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In

COMPUTER ENGINEERING

Submitted By

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This is to certify that,

Has satisfactorily carried out and completed Mini Project Work entitled,

"Titanic Survival Prediction Using Machine Learning"

It is submitted in the partial fulfillment of the prescribed syllabus of

Savitribai Phule Pune University, Pune

For the academic year 2025-2026

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Satyam K. Gawali

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DECLARATION

We are students of "Shatabdi Institute of Engineering & Research, Nashik" enrolled in the Degree in Bachelor of Engineering in Computer Engineering program, hereby declare that the mini project Report titled "Titanic Survival Prediction Using Machine Learning" is our own original work.

We further declare that all references and sources used in the preparation of this report have been duly acknowledged. This report has not been submitted for any other academic purpose, nor has it been presented elsewhere for evaluation.

We express gratitude to supervisor **Prof. P. M. Jadhav** for their invaluable guidance and support during this training period.

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ABSTRACT

The Titanic Survival Prediction project aims to build a machine learning model that predicts the likelihood of passenger survival during the RMS Titanic disaster based on demographic and socio-economic factors. Using the publicly available Titanic dataset from Kaggle, this project involves data preprocessing, exploratory data analysis, feature selection, and model training using various machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest.

The model is evaluated using performance metrics like accuracy, precision, recall, and F1-score to determine its predictive capability. The analysis identifies key features that influenced survival rates, including gender, passenger class, and family size. Developed in Python with libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn, this project demonstrates how data-driven techniques can be applied to historical datasets to uncover meaningful patterns and make reliable predictions.

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1. Aim

To build a machine learning model that predicts the survival of passengers on the Titanic based on their personal and socio-economic information.

2. Objectives

- To understand and preprocess real-world data from the Titanic dataset.
- To apply machine learning algorithms to predict survival outcomes.
- To evaluate model performance using appropriate metrics.
- To identify important features affecting passenger survival.

3. Problem Statement

The sinking of the RMS Titanic is one of history's most infamous shipwrecks. Although there was some element of luck in surviving, it is clear that certain groups of people were more likely to survive than others — such as women, children, and those in higher classes. The goal of this project is to develop a **predictive machine learning model** that estimates a passenger's likelihood of survival based on features like **age, gender, class, and family relationships**, using the **Titanic dataset** provided by Kaggle.

4. Hardware and Software Requirements

Hardware	Specification
Processor	Intel i3/i5 or above
RAM	Minimum 4 GB
Storage	200 MB free space
System	Any modern multi-core computer

Software	Details
Operating System	Windows / Linux
Programming Language	Python
IDE / Environment	Jupyter Notebook / Google Colab
Libraries Used	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

5. Algorithm / Methodology

Step 1: Data Collection

Download the Titanic dataset from Kaggle: https://www.kaggle.com/competitions/titanic/data

Step 2: Data Preprocessing

- Handle missing values in Age, Cabin, and Embarked columns.
- Convert categorical features (Sex, Embarked) into numerical form using label encoding or one-hot encoding.
- Select relevant features such as Pclass, Sex, Age, SibSp, Parch, and Fare.

Step 3: Feature Selection

Identify key features affecting survival using correlation and visualization (heatmaps, bar plots).

Step 4: Model Building

Train a **Logistic Regression** model and optionally test with other algorithms like **Decision Tree** or **Random Forest**.

Step 5: Model Evaluation

Evaluate the model using **accuracy**, **precision**, **recall**, and **confusion matrix** on test data.

6. Performance Analysis

Algorithm	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	81–83	0.84	0.74	0.79
Decision Tree	78–82	0.80	0.72	0.76
Random Forest	84–86	0.85	0.78	0.81

Observation:

The Random Forest model generally performs slightly better than Logistic Regression due to its ensemble nature. However, Logistic Regression offers simpler interpretation and faster execution, making it ideal for baseline prediction.

7. Result and Discussion

The model successfully predicts whether a passenger survived the Titanic disaster using passenger attributes. Key influencing factors are **gender**, **age**, and **passenger class (Pclass)**. Women and children from higher socio-economic classes had a higher chance of survival. The logistic regression model achieved an accuracy of around **82%**, showing strong predictive performance.

8. Conclusion

The Titanic Survival Prediction project demonstrates the power of machine learning in analyzing real-world datasets. The model effectively predicts passenger survival with good accuracy after proper preprocessing and feature selection. This project highlights the importance of data cleaning, feature encoding, and evaluation metrics in building reliable predictive models.

Build a machine learning model that predicts the type of people who survived the Titanic shipwreck using passenger data (i.e. name, age, gender, socio-economic class, etc.). Dataset Link: https://www.kaggle.com/competitions/titanic/data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
# Load the dataset (after downloading from Kaggle)
train_data = pd.read_csv('train.csv')
test_data = pd.read_csv('test.csv')

# Check first few rows
train_data.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

Next steps: Generate code with `train_data` | New interactive sheet

```
# Basic info
train_data.info()

# Check missing values
```

```
# Check missing values
train_data.isnull().sum()

# Summary statistics
train_data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	F
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

```
# Fill missing Age with median
train_data['Age'].fillna(train_data['Age'].median(), inplace=True)

# Fill missing Embarked with mode
train_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)

# Drop Cabin column
train_data.drop('Cabin', axis=1, inplace=True)

# For test data, do same
test_data['Age'].fillna(test_data['Age'].median(), inplace=True)
test_data['Fare'].fillna(test_data['Fare'].median(), inplace=True)
test_data.drop('Cabin', axis=1, inplace=True)
```

```
/tmp/ipython-input-1150473665.py:2: FutureWarning: A value is trying to be set on a c
The behavior will change in pandas 3.0. This inplace method will never work because t
```

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(


  train_data['Age'].fillna(train_data['Age'].median(), inplace=True)
/tmp/ipython-input-1150473665.py:5: FutureWarning: A value is trying to be set on a c
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(


  train_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)
/tmp/ipython-input-1150473665.py:11: FutureWarning: A value is trying to be set on a
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(


  test_data['Age'].fillna(test_data['Age'].median(), inplace=True)
/tmp/ipython-input-1150473665.py:12: FutureWarning: A value is trying to be set on a
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(


  test_data['Fare'].fillna(test_data['Fare'].median(), inplace=True)
```

```
# Strip spaces and lowercase all strings
train_data['Sex'] = train_data['Sex'].str.strip().str.lower()
test_data['Sex'] = test_data['Sex'].str.strip().str.lower()

train_data['Embarked'] = train_data['Embarked'].str.strip().str.upper()
test_data['Embarked'] = test_data['Embarked'].str.strip().str.upper()

# Fill missing values first
train_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)
test_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)

train_data['Sex'].fillna(train_data['Sex'].mode()[0], inplace=True)
test_data['Sex'].fillna(train_data['Sex'].mode()[0], inplace=True)

# Now encode
sex_encoder = LabelEncoder()
train_data['Sex'] = sex_encoder.fit_transform(train_data['Sex'])
test_data['Sex'] = sex_encoder.transform(test_data['Sex'])

embarked_encoder = LabelEncoder()
train_data['Embarked'] = embarked_encoder.fit_transform(train_data['Embarked'])
test_data['Embarked'] = embarked_encoder.transform(test_data['Embarked'])
```

```
/tmp/ipython-input-3165691540.py:9: FutureWarning: A value is trying to be set on a c
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(


  train_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)
/tmp/ipython-input-3165691540.py:10: FutureWarning: A value is trying to be set on a
The behavior will change in pandas 3.0. This inplace method will never work because t
```

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(
```

```
  test_data['Embarked'].fillna(train_data['Embarked'].mode()[0], inplace=True)
/tmp/ipython-input-3165691540.py:12: FutureWarning: A value is trying to be set on a
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(
```

```
  train_data['Sex'].fillna(train_data['Sex'].mode()[0], inplace=True)
/tmp/ipython-input-3165691540.py:13: FutureWarning: A value is trying to be set on a
The behavior will change in pandas 3.0. This inplace method will never work because t

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(
```

```
  test_data['Sex'].fillna(train_data['Sex'].mode()[0], inplace=True)
```

```
features = ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']
X = train_data[features]
y = train_data['Survived']
```

```
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state
```

```
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
RandomForestClassifier(random_state=42)
```

```
y_pred = model.predict(X_val)

print("Accuracy:", accuracy_score(y_val, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_val, y_pred))
print("\nClassification Report:\n", classification_report(y_val, y_pred))
```

```
Accuracy: 0.8212290502793296
Confusion Matrix:
 [[92 13]
 [19 55]]

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.88      0.85       105
           1       0.81      0.74      0.77        74

    accuracy                           0.82       179
   macro avg       0.82      0.81      0.81       179
weighted avg       0.82      0.82      0.82       179
```

```
test_X = test_data[features]
test_preds = model.predict(test_X)

# Save to CSV for Kaggle submission
submission = pd.DataFrame({
    'PassengerId': test_data['PassengerId'],
    'Survived': test_preds
})
submission.to_csv('titanic_submission.csv', index=False)
```

```
sns.countplot(x='Survived', hue='Sex', data=train_data)
plt.show()
```

